

Stakeholder Engagement Plan

Tokenizer for African Languages — Bridging the Tokenization Gap in African LLMs via SuperBPE —
TRI AI Saturdays, Cohort 10

This plan builds on the project's Data Card and covers two stakeholder groups whose input directly bears on decisions already documented there: the uneven per-language data coverage identified for Chichewa and Wolof, and the project's reliance on community-built resources from the African NLP research ecosystem.

Stakeholder Group 1: Chichewa and Wolof Speaker Communities

1. Who They Are and Why They Matter

Who they are. Native speakers of Chichewa and Wolof, including community members, local educators, and cultural organizations connected to these two language communities.

Why important. The project's own Data Card identifies Chichewa and Wolof as the two languages with the smallest usable corpora by word count — Chichewa at 204,640 words (well short of the project's 500,000-word working target) and Wolof at 458,166 words even after being supplemented with a second source — against corpora of over 1 million words for most other languages in the set. Any tokenizer trained on this data will most directly reflect these two languages' thinner representation, and these communities are best placed to judge whether the tokenizer's output reflects their language accurately despite that gap.

What matters most to them. Based on the concerns typically raised by speakers of lower-resource languages in AI projects: accurate representation of their language rather than being an afterthought bolted onto a project built around larger languages; visibility into why their language's data is thinner, rather than a silent quality gap; and assurance that being included in a nine-language project does not mean being treated as equally well-served when the underlying data plainly is not.

2. Potential Benefits and Risks

Potential benefits: inclusion in a functioning multi-language tokenizer at all, rather than being left out of African-language AI tooling entirely, as is common for lower-resource languages; a documented, honest account of their language's current data limitations (recorded plainly in the Data Card's Known Limitations section), which creates a concrete basis for follow-up data-collection efforts rather than an invisible gap; and a direct channel to flag tokenization errors specific to their language before the tokenizer is used in any downstream product.

Potential harms or concerns: a tokenizer that performs visibly worse for their language than for better-resourced languages in the same project, which could reinforce a sense that their language is treated as secondary even within a project explicitly built to address language-based digital exclusion; and the risk of the project citing “nine languages covered” in a way that overstates actual quality parity across those languages.

Ethical, social, and cultural considerations: dignity and self-representation are at stake here specifically because these are the two most data-scarce languages in the set, the point at which a well-intentioned

inclusion project most risks becoming tokenistic if the people affected are not consulted directly. There is also an economic dimension: any future application built on this tokenizer inherits whatever quality gap exists today, so early input from these communities has real downstream consequence, not just symbolic value.

3. Engagement Method: Participatory Design

Participatory Design is the appropriate method here because it treats community members as active contributors to the tokenizer's evaluation and refinement, not as passive subjects whose language is being processed on their behalf. This method is specifically suitable for this stakeholder group because the core concern — whether the tokenizer actually handles their language well — is something only fluent speakers can meaningfully judge; a technical quality metric like fragmentation or compression ratio cannot substitute for a native speaker reading real tokenizer output and saying whether it looks right.

In practice, this would be conducted as a small set of structured co-review sessions, ideally in partnership with an existing community organization or educator network in each language, rather than a one-off survey. Participants would be shown real example sentences in their language alongside how the tokenizer segments them, asked to flag segmentations that look wrong or that split meaningful units — a Bantu noun-class prefix in Chichewa, or a Wolof class-marker suffix — in ways that seem linguistically arbitrary. Sessions would be conducted in the participants' own language wherever possible, kept short given this is a diagnostic exercise rather than a full curriculum activity, and would explicitly walk through the Data Card's existing findings about their language's data limitations first, so participants are reacting to a fully disclosed picture rather than being asked to evaluate something presented as already complete and equal.

Stakeholder Group 2: African NLP Researcher and Language-Technology Community

1. Who They Are and Why They Matter

Who they are. Researchers, developers, and grassroots initiatives working on African-language natural language processing, including the community associated with resources this project directly draws on, such as the Masakhane initiative.

Why important. This project's Wolof corpus is directly supplemented with MasakhaNER 1.0, a dataset built and released by the Masakhane community specifically to support African-language NLP research. This project is not operating in isolation from this community's work, and future decisions — what data to add next, how to document limitations, how to license and share outputs — should continue to be checked against this community's established practice and licensing norms rather than decided independently.

What matters most to them. ethical and properly licensed data sourcing (this project uses MasakhaNER under its CC-BY-4.0-NC terms and Wikipedia under CC-BY-SA-4.0); honest, complete documentation of limitations rather than polished claims that overstate what a resource can do; and avoiding extractive research patterns where African-language data or research findings are used without engaging the researchers and communities who generated the underlying knowledge.

2. Potential Benefits and Risks

Potential benefits: a genuinely reusable, transparently documented tokenizer and data pipeline that other researchers in this space can build on directly rather than duplicating the same multi-source data collection effort from scratch; a concrete, evidenced example of the SuperBPE approach applied specifically to African-language morphology, including an honestly reported negative result at small model scale, which contributes a real data point to a small existing literature; and a project that models thorough documentation practice (data cards, honest limitation logs, disclosure of a data-hygiene issue found after initial results) that others in the community could reasonably adopt.

Potential harms or concerns: if released without proper engagement, the project could be seen as using a community-built resource (MasakhaNER) without meaningfully crediting or engaging the community that produced it; technical claims (compression gains, cross-family coverage) that are not held to the same evidentiary standard this community expects could damage trust in the work and, by association, in similar future projects.

Ethical, social, and economic considerations: this community operates largely on open, collaborative norms with limited funding relative to well-resourced commercial NLP labs; a project that draws on their prior work without reciprocal openness (releasing the tokenizer, the Data Card, and the methodology openly) would sit at odds with those norms and could be seen as extractive even if unintentionally so.

3. Engagement Method: Distributed Dialogue

Distributed Dialogue is appropriate for this group because the African NLP research community is itself geographically distributed across many countries and institutions, with established asynchronous channels (mailing lists, community forums, shared repositories) already in active use rather than a single physical location where a workshop could realistically gather representative participation. This method is specifically suitable here because the core need is not a one-time co-design session but an ongoing, low-friction way for a dispersed technical community to review and comment on documentation and methodology as it develops.

In practice, this would be conducted by sharing the project's Data Card, technical report, and eventual tokenizer release through existing community channels this research community already uses, with an explicit, direct invitation for comment on the data sourcing decisions, the documented limitations, and the methodology, rather than a silent release. Structured prompts would be included, specifically asking whether the per-language retention disparities and the Chichewa/Wolof data-scarcity findings are being characterized fairly, whether the disclosed data-hygiene issue in the evaluation methodology is being handled appropriately, and whether any known community data-sharing norms are being missed, so feedback is anchored to concrete decisions rather than open-ended commentary. Given the asynchronous nature of this method, a reasonable response window (several weeks) would be built into the release timeline rather than treating community feedback as optional or after-the-fact.

4. Influence on Project Design

Feedback from these two stakeholder groups is intended to feed back into concrete project decisions, not just be collected and set aside:

Dataset

- Chichewa and Wolof community feedback directly informs which additional data sources to prioritize sourcing next, targeting the specific gap the Data Card already documents rather than collecting more data generically.
- Researcher community feedback on data sourcing ethics and licensing informs which sources are safe to add and how to properly credit community-built resources like MasakhaNER.

Model / Tokenizer Design

- Specific segmentation errors flagged by Chichewa and Wolof speakers during participatory review sessions directly inform whether the corpus-mixing weights or vocabulary allocation need adjustment for these languages specifically, rather than assuming the current configuration generalizes evenly across all nine languages.

Outputs and Evaluation

- Community-flagged segmentation issues become concrete test cases added to the evaluation suite, so future versions are checked against real speaker-identified failure cases, not only aggregate statistical metrics like fragmentation or compression ratio.
- Researcher community feedback on methodology transparency informs what additional detail gets added to the Data Card and technical report in future revisions, including how the disclosed data-hygiene issue is ultimately resolved.

Deployment Decisions

- If Chichewa and Wolof community feedback indicates the tokenizer is not yet reliable enough for their language, this directly informs whether the project should caveat or delay recommending the tokenizer for production use in those two languages specifically, rather than presenting all nine languages as equally deployment-ready.
- Researcher community input on licensing and openness norms informs how and where the tokenizer and its underlying data pipeline are eventually released.